Clinical Guideline for:
Peri-procedural management of antiplatelet drugs

KEY POINTS

- For patients solely on clopidogrel, follow algorithm for aspirin, but, if treatment is to continue perioperatively, switch to aspirin where possible.
- ***Stopping clopidogrel or aspirin preoperatively should be considered an exception on an individual basis*** and only when undergoing procedures with high risk from bleeding or expected major bleeding complications.
- If aspirin or clopidogrel are stopped, early post-operative re-institution is important.
- Further detail can be found in the Handbook of Perioperative Medicines (available on the Preparation for Surgery section of the HUB).

INDICATIONS FOR ANTIPLATELET DRUGS

Cardiology indications for antiplatelet drugs

1. *Primary prevention of myocardial infarction(MI*)
   Primary prevention is aimed at prevention of MI in patients who have not previously suffered a myocardial event. Generally antiplatelet monotherapy with aspirin is used (or clopidogrel if aspirin allergic or intolerant). This can be stopped perioperatively with minimal risk.

2. *Secondary prevention following MI*
   Dual antiplatelet therapy (DAPT) is usually prescribed – aspirin + one of clopidogrel/prasugrel or ticagrelor.

3. *Following percutaneous coronary intervention (PCI)*
   DAPT is prescribed to prevent stent thrombosis. This is needed for 6-12 months after PCI (until endothelialisation of the stent has occurred). Exact duration is dictated by the type of stent used, **however 99% of the stents currently inserted in this Trust are drug eluting stents** – these require a minimum of 6 months DAPT to avoid stent thrombosis.

Stroke/TIA indications for antiplatelet drugs

1. There is no evidence for use of antiplatelet drugs for the primary prevention of stroke or TIA (i.e. in patients who have not previously suffered stroke or TIA).

2. Secondary prevention of stroke / TIA:
   - The standard treatment is clopidogrel 75mg daily (licenced for use in ischaemic stroke, off-label use in TIA).
   - Modified-release dipyridamole 200 mg twice daily may be used if both clopidogrel and aspirin are contraindicated or cannot be tolerated.
   - Aspirin 75mg daily may be used if both clopidogrel and modified-release dipyridamole are contraindicated or cannot be tolerated.

- Aspirin 75 mg daily with modified-release dipyridamole 200 mg twice daily may be used as an alternative to clopidogrel if not tolerated.
- Dual therapy with aspirin and clopidogrel may be initiated in secondary care for the first three months following ischaemic stroke or TIA due to severe symptomatic intracranial stenosis.
- **Bridging therapy;** there is not conclusive evidence in favour of bridging therapy although trials are currently are currently investigating the role of shorter acting intravenous antiplatelet drugs, such as tirofiban and eptifibatide. Unfractionated and low molecular weight heparin have relatively minor effects on platelets, do not prevent a thrombotic event and so are not indicated for bridging.

Peripheral arterial disease indications

- Clopidogrel 75mg daily is the preferred antiplatelet medication.
- If clopidogrel is contraindicated or not tolerated, give low dose aspirin alone.

ANTIPLATELET AGENTS IN CLINICAL PRACTICE

Cyclo-oxygenase inhibitors:
- Aspirin

P2Y12 inhibitors:
- Clopidogrel
- Prasugrel
- Ticagrelor
- Cangrelor

Phosphodiesterase-3 inhibitors:
- Dipyridamole

Glycoprotein IIb/IIIa inhibitors:
- Abciximab
- Eptifibatide
- Tirofiban

General advice

Aspirin

Aspirin permanently inactivates the platelet enzyme cyclo-oxygenase, the effect of which is only reversed by the generation of new platelets.

Aspirin is considered safe to continue peri-operatively, except for neurosurgery, and surgery involving the posterior eye chamber and medullary canal.

Stopping aspirin (secondary prevention) has been shown to significantly increase Major Adverse Cardiac Events (MACE) in 30-day peri-operative period. Continuing aspirin has not been shown to increase major bleeding events (except in POISE-2 trial, which has not been replicated).[3]

Clopidogrel

Clopidogrel has a short half-life (4 hours) but platelet inhibition is irreversible. It therefore exerts its effect for the lifespan of the platelet (approximately 7-10 days).

Discontinuation of clopidogrel is unnecessary in procedures that are not highly invasive and have a low bleeding risk or haemostasis is controllable. In general, current opinion is that clopidogrel monotherapy when taken for secondary prevention provides cardiovascular

benefit that outweighs the bleeding risk and should be continued during most procedures. Stopping clopidogrel preoperatively should be considered an exception on an individual basis and only when undergoing procedures with high risk from bleeding or expected major bleeding complications (e.g. closed space procedures transurethral prostatectomy and intraocular or intracranial procedures). One recent observational study has suggested that bleeding risk may outweigh the risk of cardiac complications but this remains to be clearly demonstrated in randomised controlled studies.[1]

The flow chart below guides decisions when the surgeon feels it is necessary to stop antiplatelet drugs.

Dipyridamole

Dipyridamole, a PDE3 inhibitor, has less antiplatelet activity than aspirin and clopidogrel. Its action ceases 24 hours after cessation. Dipyridamole need not be stopped preoperatively unless the surgical procedure is high risk (spinal surgery or other case with risk of bleeding within a confined space). Evidence regarding neuraxial blockade is limited. Sole therapy, neuraxial block is considered safe, if in combination with aspirin, delay block for 24 hours.

Newer agents

Nationally the use of prasugrel is increasing as it is more efficacious than clopidogrel in DAPT to prevent in stent thrombosis. Although use in this Trust is low patients from other Trusts are increasingly taking ticagrelor.

Prasugrel

Prasugrel has advantages over clopidogrel that include rapid onset of action, more potent platelet inhibition and fewer 'hypo-responders'. The effect is irreversible. Clinical trials suggest benefit over clopidogrel in patients undergoing percutaneous coronary intervention (PCI) following presentation with acute coronary syndrome (ACS).

The anti-platelet effect of aspirin, clopidogrel and prasugrel can be treated with platelet transfusion.

Ticagrelor

Ticagrelor is a reversible P2 Y12 platelet ADP receptor antagonist with a similar pharmacokinetic profile to prasugrel. Clinical trials have shown benefit over clopidogrel in patient presenting with ACS (irrespective of whether they underwent revascularisation).It is indicated in conjunction with aspirin in patients with ACS, or a history of MI, who are at high risk of an atherothrombotic event. Its half-life is 12 hours meaning that 3 days cessation is sufficient. As the effect is reversible, platelet transfusion may not work, as they will also be inhibited.

Ticagrelor has been shown to be eliminated using Cytosorb, a cytokine-removal filter that can be used in any haemodialysis or haemofiltration circuit[4]

Cangrelor

Currently only licensed in PCI where oral agents have not been given, this intravenous infusion has a plasma half-life of only 3-6 minutes. It is broken down in the circulation by dephosphorylation and on cessation, normal platelet function resumes within 60-90 minutes. It may herald bridging opportunities for patients in high risk situations in flow chart below.

Dual anti-platelet therapy

DAPT is required for both bare metal (4 weeks) and dug-eluting stents (6-12 months) post PCI. 4-8% of these patients will require surgery within a year of insertion. The risk of perioperative MACE post-MI is greatest in the first 4 weeks then 2-6 months, 6-12 months and least after a year.

Patients on dual antiplatelet therapy are at a higher risk of bleeding; where possible elective surgery should be delayed until the second antiplatelet can be safely stopped. When surgery cannot be delayed and the patient's thrombotic risk is high, the continuation of both agents should be discussed with the surgeon, anaesthetist and cardiologist. These patients should be considered on an individual basis. If cardiovascular thrombotic risks are considered too great to stop DAPT, consideration to bridging with intravenous cangrelor or a GIIa/IIIb inhibitor should be risk assessed.

Glycoprotein IIa/IIIb inhibitors

These drugs bind to the receptor, preventing binding of von Willebrand factor and fibrinogen. Abciximab is a F_{ab} fragment monoclonal antibody whereas tirofiban and eptifibatide are not. Their use in acute MI PCI has diminished since the increased use of P2Y12 inhibitors. Currently, the American College of Cardiology/American Heart Association recommends the use of GP IIb/IIIa inhibitors in ACS when the patient is allergic or cannot tolerate P2Y12 inhibitors (Class I) or in patients who are undergoing PCI have received P2Y12 inhibitors but at higher risk for thrombus and in those who are allergic to aspirin (Class IIa). Short duration infusions or bolus only are used. Duration of action is 4 hours thus any emergency surgery at time of MI should be delayed beyond this if possible.

Advice for vascular patients

Bypass or aortic surgery: patients should always be taking an antiplatelet agent at the time of surgery - aspirin is preferred to clopidogrel.

Carotid endarterectomy: often on dual therapy (recent stroke/TIA). Bespoke treatment as per surgeons orders in notes.

Emergency surgery on DAPT

Surgical emergency may preclude cessation of one or all anti-platelet therapy. Peri-operative platelet transfusion should be considered, appreciating that ticagrelor in particular will inhibit transfusd platelets as well, due to its reversible action. Transfusion less than 2 hours after ingestion of aspirin and 24 hours after clopidogrel may also cause inhibtion of transfused platelets.

Platelet transfusion is most effective when treating aspirin inhibition as transfused platelets act as a thromboxane donor to the aspirin-bound platelets.

Inter-individual variation of response to clopidogrel may allow some patients' platelet function to return much quicker, if they are 'non-responders'.

Platelet mapping using thromboelastography tests for platelet aggregation response to ADP and can predict degree of platelet inhibition. This can guide peri-operative risk.

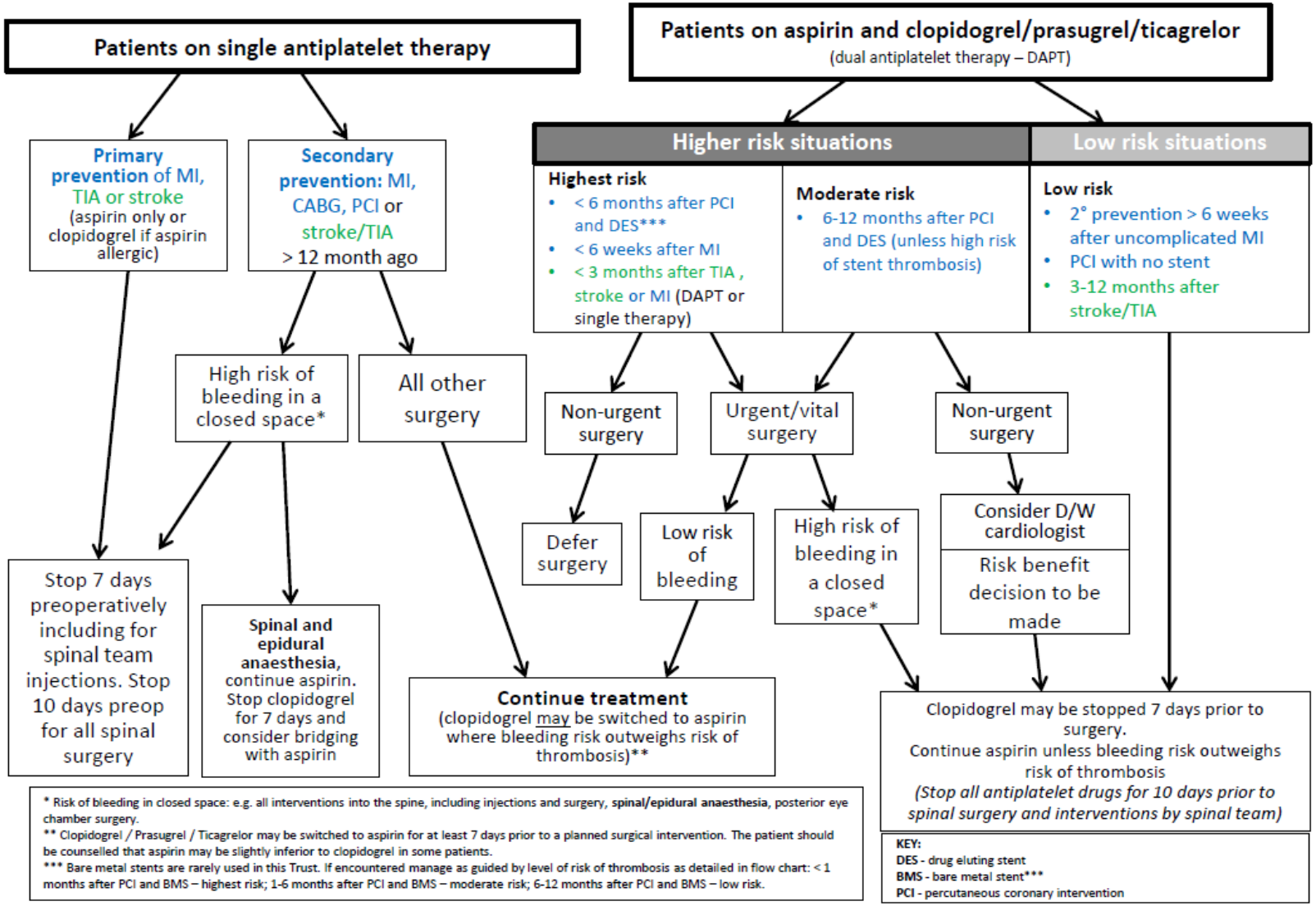
For ticagrelor, consider cytosorb membrane removal by haemodialyis or haemofiltration prior to surgery.

Summary regarding safe neuraxial blockade

Further advice on antiplatelet drugs and neuraxial anaesthesia can be found at:

neuraxial-procedure-v2-3.26.19.xlsx (stanford.edu)

Perioperative management of antiplatelet drugs for cardiology patients and patients with stroke/TIA undergoing elective surgery



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